

QueryLens - Final PRD for Antigravity

Project Goal

Build a Flask-based AI-powered analytics platform that converts English questions into safe executable SQL queries for an e-commerce database and presents results, explanations, and visualizations.

Core Stack

Flask, Python, SQLite, Olist Brazilian E-commerce Dataset, SQLCoder, SQLGlot, Plotly, HTML/CSS/JavaScript/Bootstrap.

MVP Scope

English-to-SQL, schema-aware prompting, SQL validation, correction pipeline, SELECT-only execution, results table, visualizations, explanations, REST API.

Non-Goals

No authentication, user accounts, Docker, Kubernetes, Redis, microservices, fine-tuning, multiple databases, or voice input.

Architecture

User -> Frontend -> API -> Schema Retriever -> Prompt Builder -> SQLCoder -> Validation -> Security -> Correction -> SQLite -> Formatter -> Visualization -> Response.

Security

Allow only SELECT. Reject INSERT, UPDATE, DELETE, DROP, ALTER, CREATE. Reject multiple statements. Validate before execution.

Development Order

Folder Structure, Flask Skeleton, SQLite, Frontend Shell, SQLCoder, Schema Awareness, Validation, Execution, Visualization, Explainability, Testing.

Engineering Rules

Keep implementation simple, incremental, beginner-friendly, portfolio-ready, and aligned with the PRD.

Success Criteria

User asks a business question and receives validated SQL, executed results, explanation, and chart.